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ArtiMuse: Fine-Grained Image Aesthetics Assessment with Joint Scoring and Expert-Level Understanding

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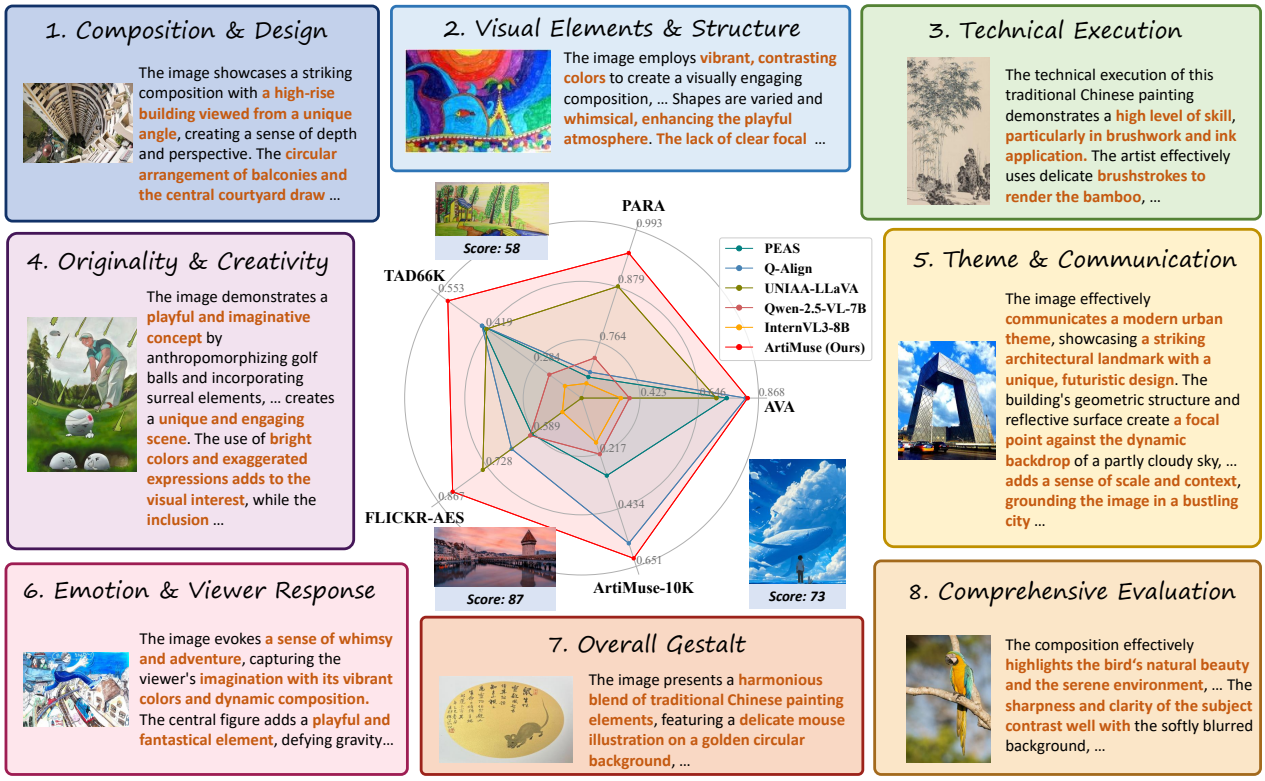


Figure 1. ArtiMuse provides granular, expert-level textual understanding results for images across 8 fine-grained aesthetic attributes. Additionally, it achieves precise image aesthetics scoring, significantly outperforming SOTA models across multiple widely-used benchmarks.

Abstract

The rapid advancement of educational applications, artistic creation, and AI-generated content (AIGC) technologies has substantially increased practical requirements for comprehensive Image Aesthetics Assessment (IAA), particularly demanding methods capable of delivering both quantitative scoring and professional understanding. Multimodal Large Language Model (MLLM)-based IAA methods demonstrate stronger perceptual and generalization capabilities compared to traditional approaches, yet they suffer from modality bias (score-only or text-only) and lack fine-grained attribute decomposition, thereby failing to support further aesthetic assessment. In this paper, we present: (1) ***ArtiMuse****, an innovative MLLM-based IAA model with Joint Scoring and Expert-Level Understanding capabilities; (2)* ***ArtiMuse-10K****, the first expert-curated IAA dataset comprising 10,000 images spanning 5 main categories and 15 subcategories, each annotated by professional experts with 8-dimensional attributes analysis and a holistic score. Both the model and dataset will be made public.*

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1. Introduction

In the era of digitalization and visual information explosion, images have become an essential medium for human beings to perceive the world, document daily life, and express emotions. From professional photography and painting to casual snapshots and sharing, images play a crucial role in conveying aesthetic values, emotional narratives, and storytelling. The advent of artificial intelligence generated content (AIGC) technologies [5, 15, 23] has further democratized visual content creation. However, this abundance of visual content also poses new challenges for quality assessment, filtering, and recommendation. While existing image quality assessment (IQA) techniques [31–33] have matured in detecting low-level degradations such as blurriness, noise, and compression artifacts, they largely focus on the technical fidelity of images and fail to capture their higher-level aesthetic attributes. Image aesthetics assessment (IAA) [7, 10, 11], which evaluates aspects such as artistic appeal, color harmony, and emotional expression, is increasingly recognized as a fundamental capability in applications including AIGC content evaluation, creative assistance, and photography education.

Despite the growing demand, current IAA methods face notable limitations. Most existing approaches rely on simplistic score predictions without capturing the inherent subjectivity, multidimensionality, and nuanced interpretations of aesthetics. Moreover, available datasets are often small in scale, coarse in granularity, and lack professionally curated annotations based on established aesthetic theories. This gap severely limits the ability of state-of-the-art multimodal large models (MLLMs) [3, 37] to understand and reason about aesthetics.

To address these challenges, we introduce **ArtiMuse**, a multimodal large language model (MLLM) for professional aesthetic understanding, together with **ArtiMuse-10K**, a meticulously curated, expert-annotated dataset. Collaborating with domain experts in aesthetics, each with 3 to over 30 years of experience, we systematically define eight explainable and fine-grained aesthetic attributes, covering aspects such as Composition & Design, Visual Elements & Structure, and Originality & Creativity, among others. Based on these attributes, we construct ArtiMuse-10K, the largest and most comprehensive fine-grained image aesthetics dataset to date, featuring both quantitative aesthetic scores and expert-written textual analyses across diverse visual domains, including graphic design, 3D design, AIGC-generated images, photography, and painting & calligraphy.

Leveraging this dataset, ArtiMuse jointly predict aesthetic scores and generate expert-level, fine-grained textual analysis, advancing aesthetic AI from mere score prediction toward holistic, interpretable reasoning. Notably, ArtiMuse achieves state-of-the-art performance across multiple widely used aesthetics benchmarks, demonstrating its robust generalization ability and superior performance in both quantitative assessment and qualitative explanation, as shown in Fig. 1 and Fig. 2.

In addition, a core technical challenge in aesthetics modeling lies in continuous score prediction using MLLMs, which are inherently designed for discrete token generation. Existing methods such as Q-Align [27] attempt to transform continuous scores into discrete ratings, and then reconstruct continuous values by weighted averaging over rating logits. However, this discretization inevitably incurs significant information loss and often leads to inaccurate predictions. To overcome this limitation, we propose a novel *Token As Score* strategy that densely maps predefined discrete tokens to continuous values. Specifically, we utilize existing tokens within the native LLM tokenizer to represent numeric values, thus eliminating the need to expand the vocabulary or retrain the tokenizer. This lightweight yet effective technique enables precise and robust modeling of continuous values within the MLLM framework, substantially improving the fidelity of aesthetics scoring.

Our main contributions can be summarized as follows:

(1) ArtiMuse-10K, a comprehensive and meticulously annotated image aesthetic assessment dataset containing 10,000 images spanning over 5 main categories and 15 subcategories. Each image is manually annotated by professional experts with detailed textual evaluations across 8 aesthetic attributes, accompanied by an overall aesthetics score. To our knowledge, this is the most extensive expert-curated dataset for aesthetic assessment to date.

(2) ArtiMuse, a novel image aesthetics assessment model, is capable of performing fine-grained expert-level textual analysis and providing accurate aesthetic scores. ArtMuse exhibits significantly superior aesthetic assessment expertise and fine-grained analysis compared to other IAA models and general-purpose MLLMs.

(3) Token As Score, which enables precise continuous aesthetics scoring in MLLMs by mapping existing tokens to numeric values, avoiding quantization loss and tokenizer changes. It offers a lightweight, effective solution for accurate and stable score prediction.

2. Related Work

2.1. Multi-modality Large Language Models

With the advancement of MLLMs [1, 3, 24, 37], their ability has expanded from basic image-text matching to understanding high-level semantic content, offering new possibilities for image aesthetics assessment. However, current MLLMs still struggle with objective evaluation, often producing overly positive and superficial judgments. Moreover, the text they generate differs significantly from the professional descriptions used by human experts, making them less suitable for high-quality automated aesthetic evaluation. Therefore, systematic fine-tuning is required to optimize and guide these models.

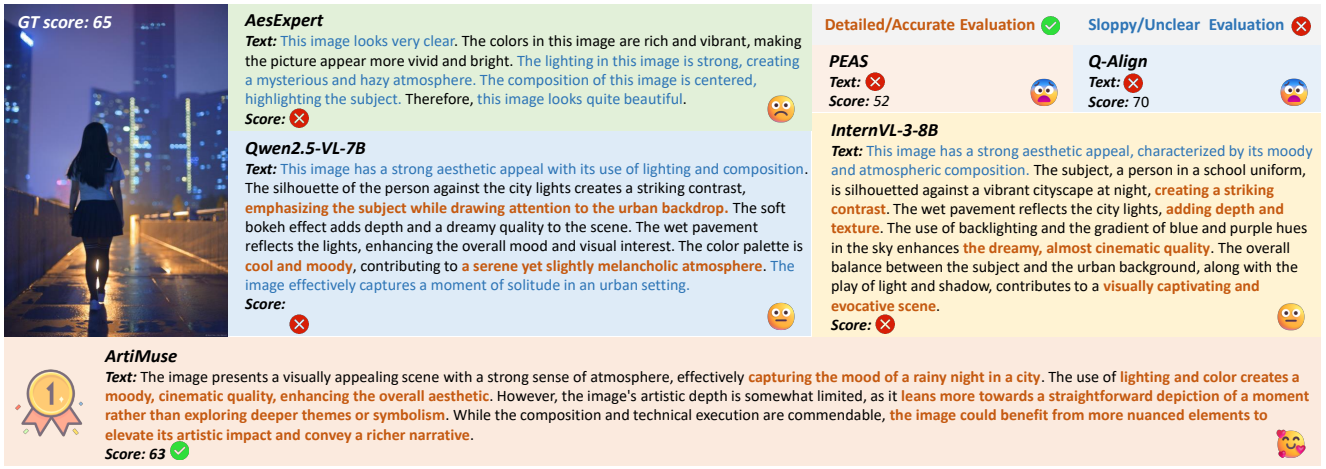


Figure 2. In comparison with existing models, ArtiMuse outperforms them by simultaneously achieving both accurate evaluation and precise aesthetics scoring in multi-dimensional assessments.

2.2. Image Aesthetics Assessment

Datasets. As summarized in Table 1, existing IAA datasets suffer from three key limitations: (1) Many [8, 13, 19, 28] offer overall aesthetics scores but lack detailed evaluative descriptions, while others [2, 14] provide only vague comments without numerical ratings; (2) Most [11, 20] focus solely on overall impressions, lacking fine-grained aesthetic attribute annotations; (3) In terms of content, datasets [13, 19, 28] are mainly photographic, with limited inclusion of artworks [2, 11, 30] and little to no AIGC or everyday scene coverage. These gaps hinder aesthetic modeling, highlighting the need for a more diverse, well-annotated benchmark.

Models. IAA models have evolved from simple regression to multimodal generative evaluation with integrated language understanding. Existing approaches fall into two categories: (1) Regression-based models (e.g., TANet [8], AesMamba [7]) predict aesthetics scores from image features but lack interpretability and generalization; (2) MLLM-based generative models leverage vision-language understanding to align better with human perception. Instruction-tuned models [26, 35] improve text generation but with limited granularity. AesExpert [10] produces expert-style descriptions but lacks score prediction. Q-Align [27] and UNIAA [36] combine text and discrete scores, yet lack fine-grained dimension-level evaluation. To overcome these gaps, we introduce ArtiMuse, a unified model delivering expert-level analysis and accurate aesthetics scoring.

3. ArtiMuse-10K Dataset

3.1. Dataset Overview

As shown in Tab. 1, ArtiMuse-10K far exceeds existing IAA datasets in diversity and granularity. It contains 10,000 images across 5 main categories (Design, AIGC, photography, etc.) with 15 fine-grained subcategories. Experts annotated each image with eight aesthetic attributes and an overall score, ensuring superior rigor and granularity.

3.2. Image Collection

Previous studies [10, 11, 27] have emphasized the importance of ensuring dataset diversity and extending domain coverage to enhance the quality and robustness of aesthetic assessment models. Building upon these insights, we construct ArtiMuse-10K, a high-quality dataset comprising 10,000 carefully curated images spanning five primary categories: Graphic Design, 3D Design, AIGC-generated images, Photography, and Painting & Calligraphy. These categories are subdivided into 15 distinct subcategories, such as Chinese Painting, Sculpture, and Daily Photography, ensuring comprehensive representation of diverse artistic expressions. The internal data samples and overall dataset composition are illustrated in Fig. 3 and Fig. 4, respectively.

Non-AIGC Images. For non-AIGC images, we collaborate with domain experts to curate professionally created artworks sourced from academic settings, including student assignments and competition entries. To ensure the dataset reflects contemporary trends, we also collect a wide range of artistic and photographic works from reputable online art and photography platforms.

AIGC Images. We utilize state-of-the-art generative models (Stable Diffusion series [23], Dreamlike Photoreal 2.0 [5], FLUX [15], etc.) to systematically produce synthetic images. We further augment this core dataset with open-source community contributions produced using comparable architectures.

3.3. Aesthetic Attributes

To establish a fine-grained annotated dataset for image aesthetics assessment, the primary task involves developing a comprehensive assessment system. Through systematic consultations with artistic experts, we have formulated a novel aesthetic assessment system. This system comprises 8 specific aesthetic attributes and an overall aesthetics score, systematically defining key dimensions of image aesthetics including Composition & Design, Visual Elements & Struc-

Table 1. A Comparison between ArtiMuse-10K dataset and existing IAA datasets.

Dataset	Main Categories	Subcategories	# Image	Score	Text Caption	# Attribute	Attribute Categories	Annotators
AVA [19]	Photography	–	255,528	✓	✗	–	–	Non-Experts
AADB [13]	Photography	–	10,000	✓	✗	–	–	Non-Experts
FLICKR-AES [22]	Photography	9 Categories	40,499	✓	✗	–	–	Non-Experts
SPAQ [6]	Photography	–	111,125	✓	✗	–	–	Non-Experts
KonIQ-10K [9]	Photography	–	10,073	✓	✗	–	–	Non-Experts
ArtEmis [2]	Painting	–	81,446	✗	✓	1 Attribute	Emotional Analysis	Non-Experts
RPCD [20]	Photography	–	73,965	✓	✓	1 Attribute	Overall Comment	Non-Experts
PARA [28]	Photography	–	31,229	✓	✗	–	–	Non-Experts
TAD66K [8]	Painting, Photography	–	66,000	✓	✗	–	–	Non-Experts
Impressions [14]	Photography	–	1,440	✗	✓	3 Attributes	Description, Perception, Evaluation	Non-Experts
BAID [30]	Painting	–	60,337	✓	✗	–	–	Non-Experts
APDDv2 [11]	Painting	3 Categories	10,023	✓	✓	1 Attribute	Overall Comment	Professional Experts
ArtiMuse-10K (Ours)	**Graphic Design, 3D Design, AIGC, Photography, Painting & Calligraphy**	**15 Detailed Categories**	**10,000**	✓	✓	**8 Attributes**	**Fine-grained Attributes (Composition & Design, Technical Execution, etc.)**	**Professional Experts**

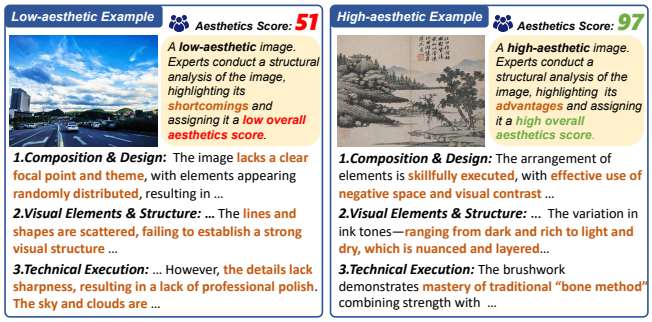


Figure 3. Data examples in ArtiMuse-10K.

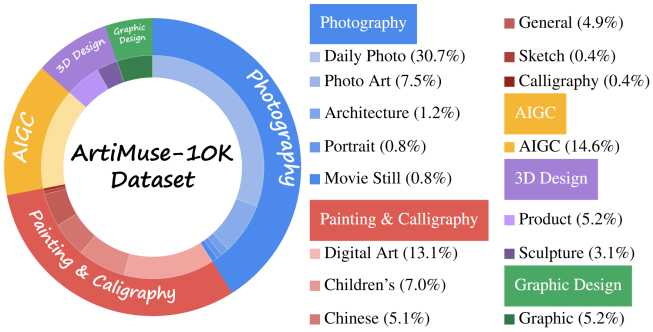


Figure 4. Composition of ArtiMuse-10K.

ture, Technical Execution, Originality & Creativity, Theme & Communication, Emotion & Viewer Response, Overall Gestalt and Comprehensive Evaluation. Notably, our system is content-agnostic and universally applicable to image types from natural to AIGC.

3.4. Human Annotations

Based on the predefined aesthetic attributes, we invite professional experts to meticulously annotate images in the ArtiMuse-10K dataset. We collaborate with domain experts whose professional experience spans a broad spectrum, ranging from at least three years to over three decades, including distinguished authorities in the field. The entire annotation process is illustrated in Fig. 5 as Type 3: Professionally Selected Images. Each image in ArtiMuse-10K is ultimately annotated with textual analysis describing eight distinct aesthetic attributes and an overall aesthetics score. Our annotation framework enhances dataset quality and model performance by integrating multi-dimensional aesthetic attributes for fine-grained visual analysis, expert-curated scores for reliable aesthetic assessment, and rich semantic annotations for improving training robustness.

4. Methodology

4.1. Dataset Collection & Processing

Richer data sources and more meticulous manual annotations are crucial for enhancing dataset quality. In addition to the ArtiMuse-10K dataset, we carefully curate over 350,000 high-quality annotated images from existing datasets, including APDDv2 [11], Impressions [14] and so on.

Aesthetic Caption Quality. We place particular emphasis on the aesthetic caption quality. Our selection criteria prioritize datasets that include valuable aesthetic-related captions such as aesthetics scores, comprehensive textual analyses, and aesthetic attribute tags. These captions are subsequently utilized in the annotation generation phase to enhance dataset quality. Ultimately, the collected captions come from annotators ranging from domain experts with **3–30 years** of professional experience to community enthusiasts and everyday users, ensuring diversity in perspectives and judgments.

Aesthetic Quality Diversity. Our collection specifically incorporates images with varying aesthetic qualities, including intentionally retained lower-quality samples, to address both dataset diversity requirements and mitigate the prevalent preference bias observed in contemporary LLMs. This carefully balanced composition strategy enhances model training through controlled inclusion of suboptimal visual materials, thereby improving discriminative capabilities in aesthetic assessment.

4.2. Annotation Generation

The Annotation Generation stage aims to enrich the dataset with detailed descriptive and evaluative annotations, illustrated in Fig. 5. This process involves creating distinct annotation types based on the available information for each image. **Type 1:** For images with only score caption, we leverage this global quality assessment to generate holistic analyses. We design a prompt to guide the MLLM in



Figure 5. Overview of ArtiMuse. ArtiMuse encompasses a multi-stage pipeline spanning data collection & processing, annotation generation, and model training, enhancing its text evaluation capabilities and score assessment proficiency across multiple dimensions.

producing a comprehensive evaluation based on predefined aesthetic attributes, while incorporating both the score and visual input. **Type 2:** For images with partial text captions containing specific aesthetic descriptions, we employ a prompt to instruct the MLLM to generate fine-grained evaluations. For each image, the model produces a structural analysis across 8 aesthetic attributes, utilizing both the textual and visual inputs. **Type 3:** For professionally selected images, we engage experts to conduct structural analysis based on pre-defined aesthetic attributes, along with providing an overall aesthetics score.

Importance of Manual Annotations. Although MLLMs demonstrate strong aesthetic evaluation capabilities, our empirical analysis reveals a systematic bias: they tend to generate overwhelmingly positive assessments regardless of the actual image quality, as shown in Fig. 7. This positivity bias leads to annotations that poorly reflect true aesthetic merit. To address this limitation, we incorporate professional human evaluations to provide balanced and reliable ground-truth annotations.

4.3. Training Strategy

ArtiMuse is built on InternVL-3-8B [37]. We modify the dynamic resolution strategy to a fixed-resolution approach while retaining the remaining components. The training process consists of two distinct phases: text pretraining and score fine-tuning, as illustrated in Fig. 5. In both stages, we jointly train the vision encoder, MLP, and LLM components, with the LLM undergoing LoRA-based fine-tuning. The ArtiMuse uses common GPT loss [21], i.e. minimizing the cross-entropy loss between the predicted logits and target tokens.

Text Pretrain. The text pretraining phase utilizes our complete collected image dataset, where each image is paired with its corresponding aesthetic analysis caption generated during the annotation generation stage. This phase aims to equip the model with accurate structural aesthetic anal-

ysis capabilities while largely preserving the MLLM's pretrained knowledge. To achieve this balance, we apply LoRA fine-tuning specifically to the LLM component.

Score Finetune. After establishing foundational aesthetic understanding through pretraining, we proceed to score fine-tuning. In this phase, we convert each image's overall aesthetics score into a specialized scoring token designed exclusively for aesthetics scoring, which then serves as the training caption. Inspired by previous works [16, 27, 36], we propose a novel score prediction strategy called *Token As Score*, which eliminates the need for vocabulary expansion or tokenizer retraining. Specifically, we designate 101 existing tokens as `[Aes_Score_Token]`s, each corresponding to integer scores ranging from 0 to 100. We select tokens that are concise and inherently carry ordinal semantic information from the vocabulary. In our implementation, we employ twin-letter combinations as tokens (e.g., Score 1 is represented as `[Aes_Score_Token_1]`, where the actual token is `ab`. See Appendix for more details). During data preprocessing, we first normalize aesthetics scores to the [0,100] range and then map them to their corresponding tokens. This methodology enables the construction of training data where continuous scores are discretized into token representations. The model is subsequently fine-tuned to predict these discrete tokens. During inference, we convert the predicted tokens back to their numerical values, and the final aesthetics score is derived by computing the expectation over the probability distribution of all possible score tokens. Specifically, we denote l_i and p_i for logits and probability of `[Aes_Score_Token_i]`, the final aesthetics score S_{Aes} is compute as:

$$S_{\text{Aes}} = \sum_{i=0}^{100} i \times p_i = \sum_{i=0}^{100} i \times \frac{e^{l_i}}{\sum_{j=0}^{100} e^{l_j}} \tag{1}$$

Why Token As Score? Current approaches for scoring with MLLMs primarily fall into two categories: (1) directly prompting the LLM to output scores as text (*Text As Score*), or (2) predefining discrete levels corresponding to specific score intervals and computing the final score based on the model's predicted token distribution (*Level As Score*). Previous works [16, 27, 36] demonstrate that directly generating scores as text leads to severe hallucination issues. Thus, we adopt the Token As Score approach and investigate the impact of token granularity on model performance. Fig. 6 presents a comparison of these score prediction methods.

Maintaining Text Ability. A widely recognized challenge in IAA and IQA tasks is that MLLMs often struggle to simultaneously preserve their textual understanding and scoring capabilities [32, 33]. Since the training data in the score fine-tuning phase is significantly more monotonous than in text pretraining, full fine-tuning of the LLM can easily degrade its structural aesthetic analysis ability. To mitigate this issue while maintaining proficiency, we employ LoRA-based fine-tuning for the LLM, enabling the model to retain both linguistic and scoring capabilities.

5. Experiments

5.1. Implementation Details

In our experiments, we adopt InternVL-3-8B [37] as the base model initialized with its pretrained weights. During text pretraining, we implement a batch size of 128 and learning rate of $4e-5$ with a cosine annealing schedule [18], training for one epoch to balance convergence with prior knowledge preservation. For the score fine-tuning, we maintain the batch size at 128 while adjusting the learning rate to $2e-5$ across 2 training epochs. We maintain identical configurations across all experiments, with all training conducted on 4 * NVIDIA A100 80GB GPUs. Text pretraining completes within 5 hours, while score fine-tuning converges efficiently in just 10 minutes on ArtiMuse-10K and 4 hours on AVA (2M-scale dataset).

5.2. Structural Aesthetic Analysis

Judging by MLLM. To evaluate current models' ability of structural aesthetic analysis, we design a judgement framework leveraging the superior comprehension power of MLLM. An image is presented to both experts and various models to generate aesthetic analysis on 8 aesthetics attributes. Then a judging MLLM selects which model performs best across each attribute, using the human expert's description as a reference. Tab. 2 show ArtiMuse outperforms other models across 8 aesthetic attributes, demonstrating superior aesthetic analysis capability.

Judging by Human. In addition, we conduct a user study where participants are asked to compare and vote for the model they perceive as producing higher-quality aesthetic analysis. The proportion of selections for each model, presented as Human Rate in Table 2, demonstrates that our approach achieves a significantly higher preference rate compared to other methods.

Qualitative Comparison. Fig. 7presents a systematic evaluation of aesthetic analysis performance across different models. Our approach demonstrates consistent superiority in analyzing both natural and AIGC images, with particular strengths in identifying key aesthetic elements such as compositional cohesion and characteristic AIGC artifacts. More results are provided in the Appendix.

5.3. Aesthetics Scoring

Comparison across Multiple Image Aesthetics Scoring Datasets. We evaluate the performance of ArtiMuse against other models across multiple Image aesthetics scoring datasets. For models unable for test (TANet [8], AesMamba [7], UNIAA-LLaVA [36], Next Token Is Enough [16]), we directly adopt the test results reported in their original papers. For models with unclear training protocols or those trained on general scenarios (MUSIQ [12], VILA [17], mPLUG-Owl2 [29], ShareGPT-4V [4], Qwen-2.5-VL-7B [3], InternVL3-8B [37], Q-Instruct [25], PEAS [34]), we test their official released

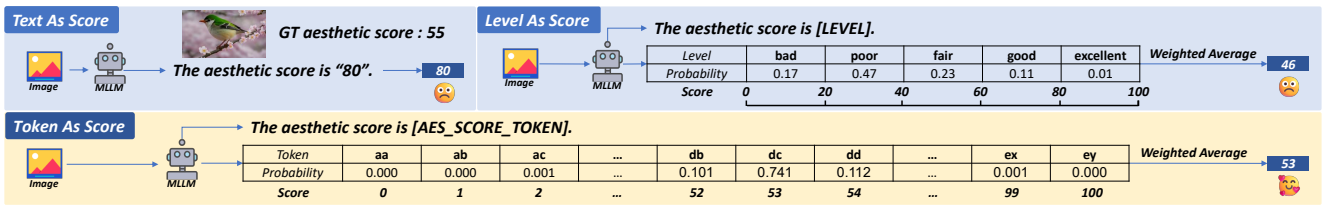


Figure 6. Comparison of score prediction methods. Token As Score features a more rational design and delivers more precise results.

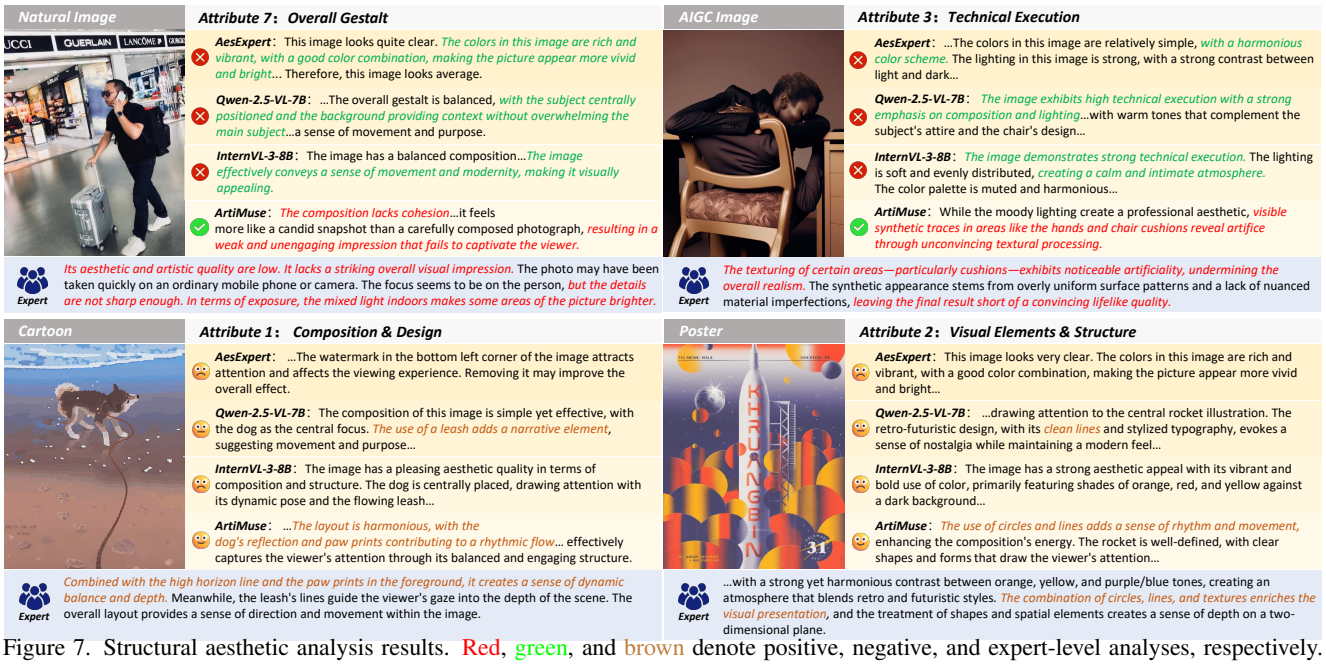


Figure 7. Structural aesthetic analysis results. Red, green, and brown denote positive, negative, and expert-level analyses, respectively. Unlike prior models, ArtiMuse uniquely diagnoses flaws in low-aesthetic images and professionally evaluates high-aesthetic ones.

models. Both Q-Align [27] and our proposed model are fine-tuned on each target dataset. As shown in Tab. 3, ArtiMuse demonstrates superior performance, achieving nearly the highest metrics across all datasets. Notably, it outperforms other models by over 0.05 PLCC on the PARA [28] and ArtiMuse-10K datasets, demonstrating its accurate aesthetics scoring capability.

Table 2. The selection rates of different models. For the first 8 aesthetic attributes, evaluations are performed by Gemini-2.0-flash, while Human Rate is provided by volunteer participants. Results are reported for AesExpert [10], Qwen-2.5-VL-7B (QwenVL) [3], InternVL3-8B (InternVL) [37], and our ArtiMuse.

Attribute	AesExpert	QwenVL	InternVL	ArtiMuse
Composition & Design	0.0%	12.7%	10.4%	**76.9%**
Visual Elements & Structure	0.0%	19.3%	16.5%	**64.2%**
Technical Execution	0.0%	9.9%	10.4%	**79.7%**
Originality & Creativity	0.0%	13.7%	8.5%	**77.8%**
Theme & Communication	0.9%	17.5%	24.1%	**58.5%**
Emotion & Viewer Response	0.0%	17.5%	24.1%	**58.5%**
Overall Gestalt	0.0%	14.6%	9.4%	**75.9%**
Comprehensive Evaluation	0.0%	17.5%	10.8%	**71.7%**
Attributes Avg.	0.1%	14.3%	14.5%	**71.1%**
Human Rate	1.5%	11.5%	19.2%	**67.8%**

Generalization Ability. We compare the generalization capabilities of ArtiMuse and Q-Align, the top-performing baseline model in comparison. Both models are fine-tuned solely on the largest AVA dataset [19] and subsequently evaluated on out-of-distribution datasets without additional adaptation. As presented in Table 3, ArtiMuse consistently achieves superior performance over Q-Align across all benchmark datasets. Remarkably, ArtiMuse's zero-shot transfer performance exceeds that of specialized IAA models, highlighting its exceptional generalization ability.

Discussion of Image Aesthetics Scoring Datasets. Prior work [16, 27, 33, 36] has consistently demonstrated that IAA remains a challenging task due to the subjective nature of aesthetic perception and the substantial distributional shifts across different datasets. Our results in Tab. 3 further corroborate this observation: while achieving strong performance on fine-tuned datasets, models often fail to generalize to unseen data.

5.4. Ablation Studies

Datasets Variants. We conduct 4 experiments (a)-(c) to systematically validate the contribution of each dataset component, as shown in Tab. 4. The results demonstrate consistent performance drop when any component is removed, with the most significant drop occurring upon exclusion of the Images w/ score caption subset, for the subset's inclusion of data from AVA. Results show dataset composition critically affects model performance.

Training Strategy. Comparative analysis between (d) and (h) reveals that full fine-tuning significantly impacts model

Table 3. Comparison on aesthetics scoring. The best and second-best performances are highlighted in red and blue, respectively. † Results are taken directly from original papers as these models cannot be tested. * Results are trained only on AVA to compare the generalization ability. For models without scoring capability, we prompt them to directly output scores as text for evaluation.

Model	AVA [19]		PARA [28]		TAD66K [8]		FLICKR-AES [22]		ArtiMuse-10K	
	SRCC	PLCC	SRCC	PLCC	SRCC	PLCC	SRCC	PLCC	SRCC	PLCC
Traditional Models										
MUSIQ [12]	0.818	0.819	0.488	0.461	0.407	0.434	0.533	0.569	0.249	0.230
TANet [8] †	0.758	0.765	–	–	0.513	0.531	–	–	–	–
VILA [17]	0.776	0.775	0.651	0.658	0.418	0.444	0.616	0.645	0.273	0.268
AesMamba [7] †	0.774	0.769	0.936	0.902	0.511	0.483	–	–	–	–
MLLMs for General-Purpose Applications										
mPLUG-Owl2 [29]	0.206	0.211	0.376	0.372	0.089	0.106	0.382	0.359	0.159	0.145
ShareGPT-4V [4]	0.213	0.199	0.509	0.417	0.097	0.091	0.335	0.289	0.076	0.057
Qwen-2.5-VL-7B [3]	0.391	0.371	0.721	0.743	0.240	0.242	0.621	0.578	0.256	0.179
InternVL3-8B [37]	0.364	0.332	0.667	0.693	0.203	0.191	0.553	0.459	0.187	0.157
MLLMs for Image Aesthetics Assessment										
Q-Instruct [25]	0.318	0.338	0.569	0.724	0.122	0.159	0.259	0.299	-0.045	-0.056
PEAS [34]	0.748	0.748	0.686	0.700	0.415	0.444	0.577	0.613	0.306	0.293
Q-Align [27]	0.822	0.817	0.913	0.888	0.501	0.531	0.798	0.818	0.551	0.573
UNIAA-LLaVA [36] †	0.713	0.704	0.864	0.895	0.411	0.425	0.724	0.751	–	–
Next Token Is Enough [16] †	0.828	0.825	–	–	0.413	0.444	–	–	–	–
ArtiMuse (Ours)	**0.827**	**0.826**	**0.936**	**0.958**	**0.510**	**0.543**	**0.814**	**0.837**	**0.614**	**0.627**
Comparison of Generalization Ability										
Q-Align *	0.822	0.817	0.694	0.711	0.417	0.445	0.643	0.664	0.337	0.320
ArtiMuse (Ours) *	0.827	0.826	0.697	0.725	0.419	0.451	0.647	0.676	0.395	0.376

Table 4. Ablation studies comparing dataset variants and training methods, with SRCC and PLCC reported on AVA.

Exp.	Images w/ Score Caption	Images w/ Partial Text Caption	Professionally Selected Images	Training Strategies	Score Prediction	SRCC	PLCC
(a)	✓	✓	–	LLM LoRA / 2-Stage Training	Token As Score (100 Tokens)	0.824	0.825
(b)	–	✓	✓	LLM LoRA / 2-Stage Training	Token As Score (100 Tokens)	0.621	0.627
(c)	✓	–	✓	LLM LoRA / 2-Stage Training	Token As Score (100 Tokens)	0.825	0.824
(d)	✓	✓	✓	LLM Full-finetune / 2-Stage Training	Token As Score (100 Tokens)	0.816	0.814
(e)	✓	✓	✓	LLM LoRA / Joint Training	Token As Score (100 Tokens)	0.821	0.820
(f)	✓	✓	✓	LLM LoRA / 2-Stage Training	Text As Score	0.820	0.819
(g)	✓	✓	✓	LLM LoRA / 2-Stage Training	Level-As-Score (Q-Align)	0.820	0.818
(h)	✓	✓	✓	LLM LoRA / 2-Stage Training	Token As Score (100 Tokens)	**0.827**	**0.826**

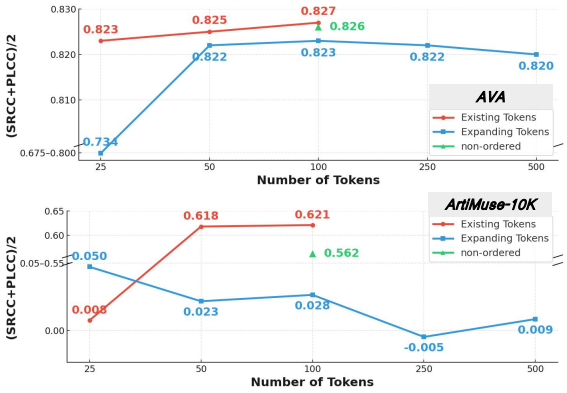


Figure 8. Token As Score Explorations on AVA & ArtiMuse-10K.

performance, primarily due to the loss of fundamental aesthetic priors acquired during the text pretraining phase. This finding is further substantiated by the comparison between (e) and (h), which conclusively demonstrates the effectiveness of our proposed 2-stage training paradigm. The results indicate that preserving pretrained text representations while adapting to score prediction tasks yields superior performance compared to end-to-end joint training approaches.

Score Prediction. We systematically examine score prediction strategies in (f)–(h). Compared with (h), (f) demonstrates suboptimal performance, confirming that Text As Score suffers from severe hallucination issues. Exp. (g) adopts Level As Score from Q-Align but performs worse than (h), as predicting with only 5 discrete levels lacks granularity and the level words can be further decomposed into tokens, introducing noise. Interestingly, (f) is comparable to (g), highlighting the importance and effectiveness of text pretraining in our pipeline. We further investigate Token As Score strategies (Fig. 8), observing that too few tokens lack sufficient granularity, whereas too many introduce excessive complexity. Moreover, expanding tokens (newly added to the vocabulary) or using non-ordered tokens consistently degrades performance. Guided by these findings, we adopt 100 existing tokens as the final setting, with further details provided in the Appendix.

6. Conclusion

We introduce ArtiMuse-10K, a large expert-annotated dataset for image aesthetics assessment (IAA), and ArtiMuse, the first model to achieve expert-level textual evaluation and precise aesthetics scoring. We further propose Token As Score, a lightweight yet effective method for continuous score prediction in MLLMs. Together these contributions will advance the field of IAA by providing more comprehensive dataset, more superior model, and more efficient scoring paradigm.

Limitations. The current model is limited to analyzing, and is unable to provide professional aesthetic enhancement recommendations, which will be addressed in future work.

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